# Regression Report - Ridge

Validation: k_folds · generated by ml_pipe

## 1. Executive Summary

| | |
|---|---|
| **model** | Ridge |
| **validation** | k_folds |
| **r2** | 0.4983 |
| **mae** | 44.2742 |
| **mse** | 2975.2465 |
| **rmse** | 54.5458 |
| **fit + predict time** | 0.008 s |

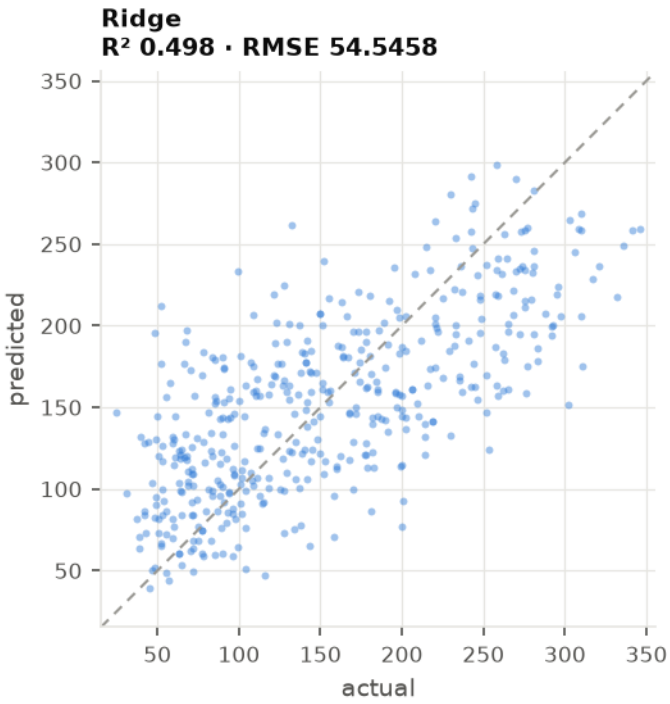
## 2. Experiment Configuration

| | |
|---|---|
| **alpha** | 0.01 |
| **copy_X** | True |
| **fit_intercept** | True |
| **max_iter** | None |
| **positive** | False |
| **random_state** | None |
| **solver** | 'auto' |
| **tol** | 0.0001 |

# 3. Regression Analysis

## Actual vs predicted

id: reg_actual_vs_predicted · model(s): Ridge

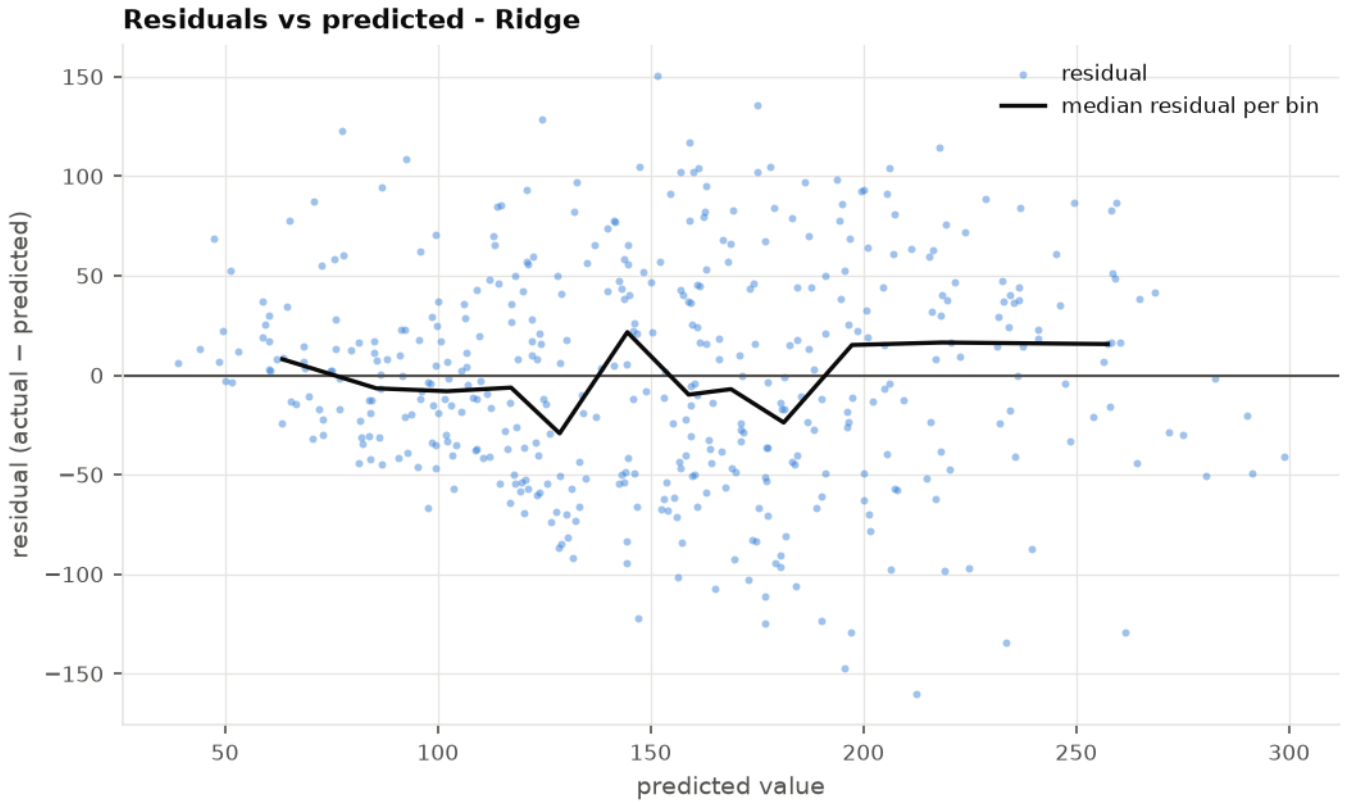



**What it shows.** Each out-of-sample prediction against its true value, one panel per model on identical axes. Points on the dashed identity line are perfect; a cloud flatter than the line shows the model shrinks toward the mean, and a widening cloud shows errors that grow with the target.

| | |
|---|---|
| **Ridge** | R² 0.4983, RMSE 54.5458 |

## Residuals vs predicted values

id: reg_residuals · model(s): Ridge

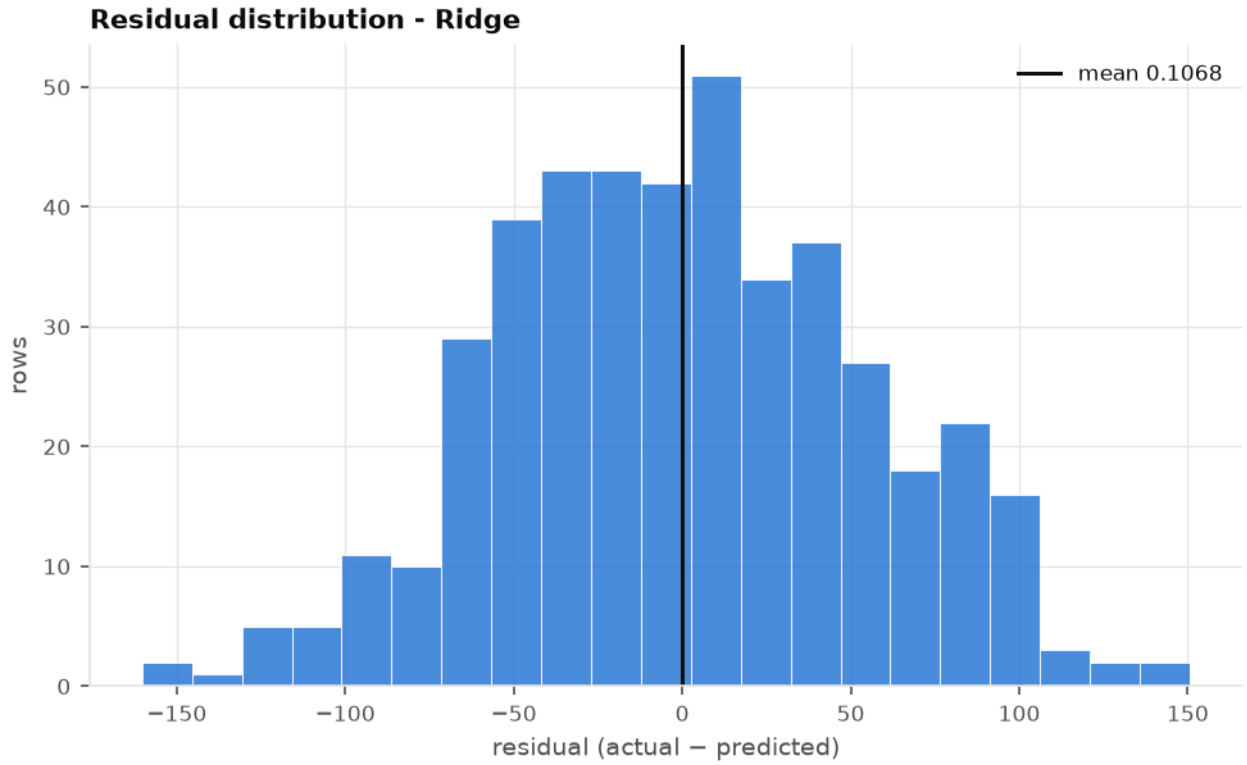



**What it shows.** Residuals of the best model against its predictions, with the median residual per prediction bin. Ideally a flat, even band around zero. A curved median line means a missed non-linearity; a funnel shape means the error grows with the prediction (heteroscedasticity).

| | |
|---|---|
| **mean residual (bias)** | 0.1068 |
| **residual SD, lower / upper half of predictions** | 47.0351 / 61.1301 |

## Residual distribution

id: reg_residual_distribution · model(s): Ridge

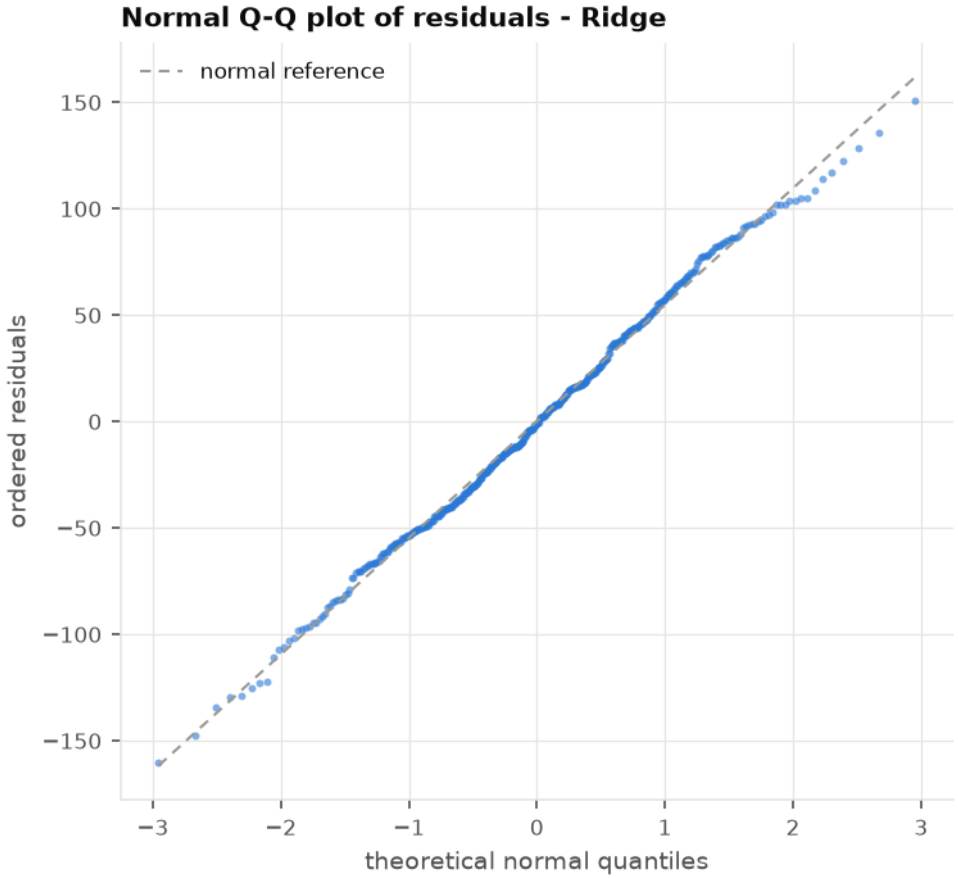



**What it shows.** Histogram of the best model's residuals. Centred on zero means no overall bias; skew or a long tail shows the model systematically under- or over-predicts some rows.

| | |
|---|---|
| **mean** | 0.1068 |
| **standard deviation** | 54.6075 |
| **skewness** | 0.0037 |
| **rows** | 442 |

## Normal Q-Q plot of residuals

id: reg_qq · model(s): Ridge

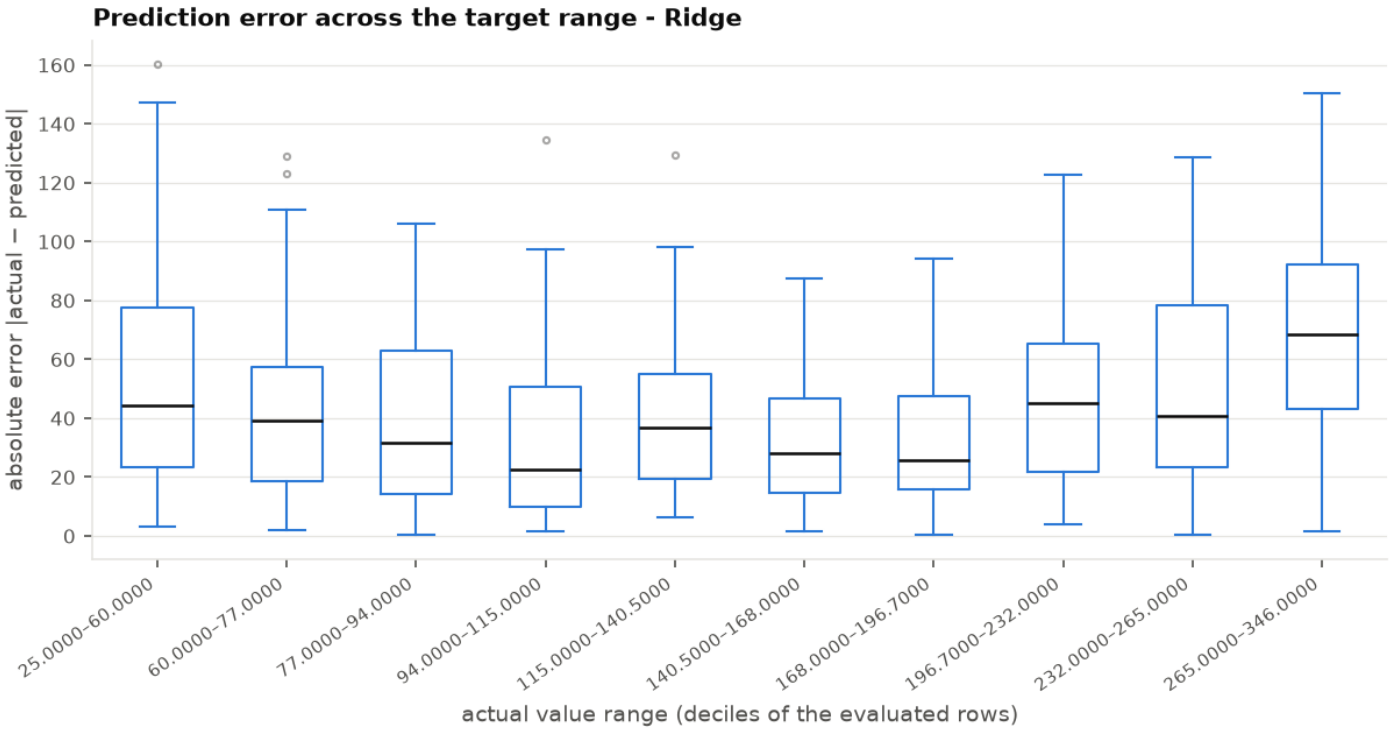



**What it shows.** Residual quantiles against those of a normal distribution. Points on the line = normally distributed errors; ends bending away = heavier tails than normal, so intervals that assume normal errors would be too narrow. A diagnostic, not a requirement of the models.

| | |
|---|---|
| **probability-plot correlation r** | 0.9986 |

## Error by target range

id: reg_error_by_range · model(s): Ridge



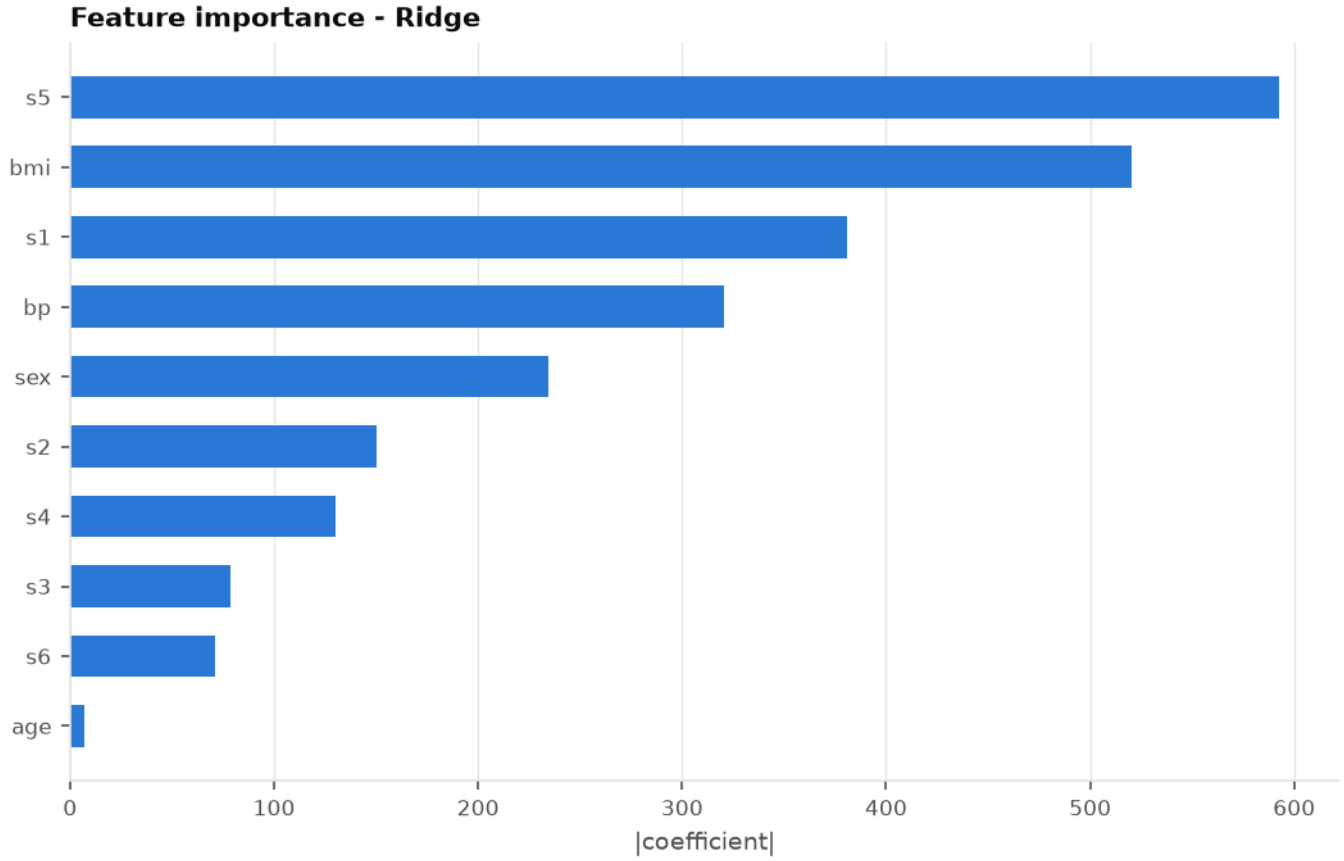

**What it shows.** Absolute errors of the best model grouped by deciles of the true value. Shows whether the model is reliable across the whole range or fails at the extremes (often where training data is scarce).

| | |
|---|---|
| **MAE** | 44.2742 |
| **worst range (median \|error\|)** | 265.0000–346.0000 (68.3922) |

# 4. Feature Analysis

## Feature importance - Ridge

id: feature_importance:Ridge · model(s): Ridge




**What it shows.** The importance the fitted model itself assigns to each input: feature_importances_ for tree ensembles, the absolute coefficient for linear models. Shows what the model relies on, not causal effects; correlated features share importance. For a model-agnostic, held-out measure use FeatureResearch.permutation_importance().

Inputs were not standardized: coefficient sizes depend on each feature's units.

| | |
|---|---|
| **s5** | 592.3480 |
| **bmi** | 520.5886 |
| **s1** | 380.6071 |
| **bp** | 320.5171 |
| **sex** | 234.5498 |
| **s2** | 150.4847 |
| **s4** | 130.3125 |
| **s3** | 78.5893 |
| **s6** | 71.1348 |
| **age** | 7.1975 |

# 5. Visualization Index

Every visualization attempted for this report: 6 generated, 0 skipped, 0 failed.

| # | visualization | section | status | time (s) | detail |
|---|---|---|---|---|---|
| 1 | Actual vs predicted | Regression Analysis | SUCCESS | 0.08 | included in the report |
| 2 | Residuals vs predicted values | Regression Analysis | SUCCESS | 0.11 | included in the report |
| 3 | Residual distribution | Regression Analysis | SUCCESS | 0.10 | included in the report |
| 4 | Normal Q-Q plot of residuals | Regression Analysis | SUCCESS | 0.11 | included in the report |
| 5 | Error by target range | Regression Analysis | SUCCESS | 0.16 | included in the report |
| 6 | Feature importance - Ridge | Feature Analysis | SUCCESS | 0.09 | included in the report |